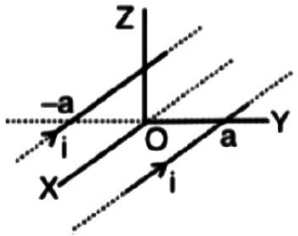
PART-1 : PHYSICS

SECTION-I (i)

1) Two long thin, parallel conductors carrying equal /currents in the same direction are fixed parallel to the x-axis, one passing through y = a and the other through y = –a. The resultant magnetic field due to the two conductors at any point is B. Which of the following are correct ?



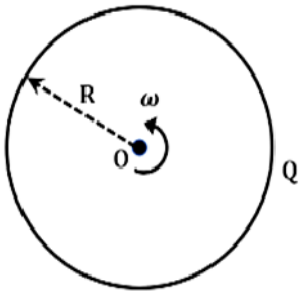
(A) B = 0 for all points on the x-axis.

(B) At all points on the y-axis, excluding the origin, B has only a z-component

(C) At all point of the z-axis, excluding the origin, B has only a y-component.

(D) B cannot have an x-component.

2) A nonconducting disc having uniform positive charge Q, is rotating about its axis with uniform



angular velocity ω. The magnetic field at the centre of the disc is-

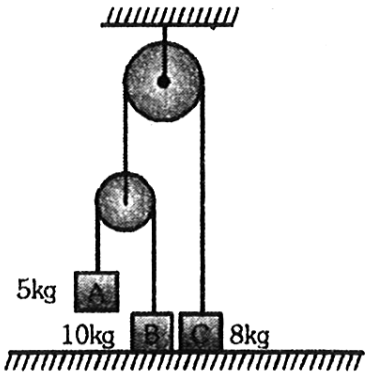
(A) directed outwards

(B) having magnitude $\frac{\mu_0 Q\omega}{4\pi R}$

(C) directed inwards

(D) having magnitude $\frac{\mu_0 Q\omega}{2\pi R}$

3) In the following arrangement the system is initially at rest. The 5 kg block is now released. Assuming the pulleys and string to be massless and smooth, the acceleration of block is



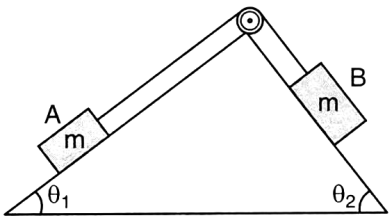
(A) $a_A = \frac{g}{7}$

(B) $a_B = 0\ m/s^2$

(C) $a_C = \frac{g}{14}$

(D) $2a_C = a_A$

4) In Fig., a body A of mass m slides on a plane inclined at angle θ_1 to the horizontal and μ_1 is the coefficient of friction between A and the plane. A is connected by a light string passing over a frictionless pulley to another body B, also of mass m, sliding on a frictionless plane inclined at angle θ_2 to the horizontal. Which of the following statement(s) is (are) true ?



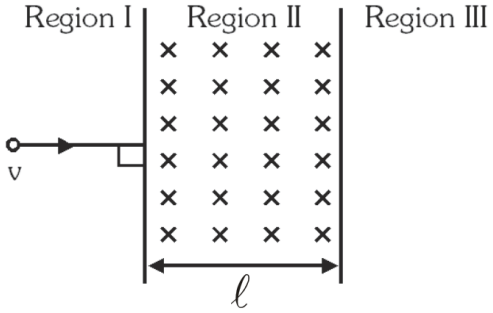
(A) A will never move up the plane

(B) A will move up the plane if $\mu < (\sin\theta_2 - \sin\theta_1)/\cos\theta_1$

(C) For A to move up the plane, θ_2 must always be greater than θ_1

(D) B will always slide down with constant speed

5) A particle of mass m and charge q, moving with velocity v enters region II normal to the boundary as shown in the figure. Region II has a uniform magnetic field B perpendicular to the plane of the paper. The length of the Region II is □. Choose the correct choice (s).



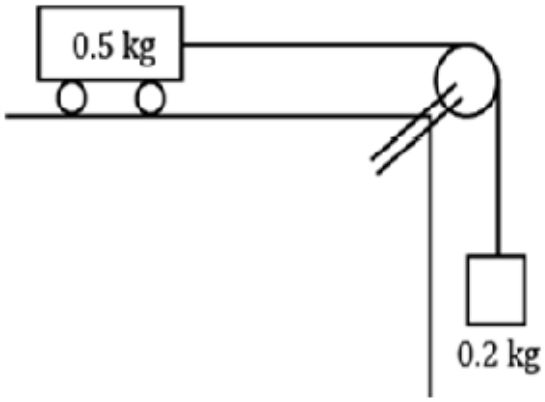
(A) The particle enters Region III only if its velocity $v > \frac{q\ell B}{m}$

(B) The particle enters Region III only if its velocity $v < \frac{q\ell B}{m}$

(C) Time spent by particle in Region II is $\frac{\pi m}{qB}$ when $v < \frac{q\ell B}{m}$

(D) Time spent in Region II is same for any velocity v as long as the particle returns to Region I

6) A cart of mass 0.5 kg is placed on a smooth surface and is connected by a string to a block of mass 0.2 kg. At the initial moment the cart moves to the left along a horizontal plane at a speed of 7



m/s. (Use g = 9.8 m/s^2)

(A) The acceleration of the cart is $\frac{2g}{7}$ towards right.

(B) The cart comes to momentary rest after 2.5 s.

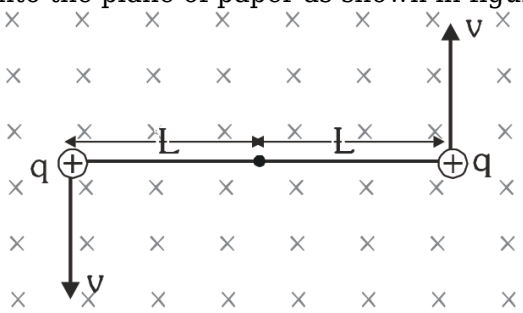
(C) The distance travelled by the cart in the first 5s is 17.5 m.

(D) The velocity of the cart after 5s will be same as initial velocity.

SECTION-I (ii)

Common Content for Question No. 1 to 2

Two charge particles each of mass 'm', carrying charge +q and connected with each other by a massless inextensible string of length 2L are describing circular path in the plane of paper, each with speed $v = \frac{qB_0L}{m}$ (where B^0 is constant) about their centre of mass in the region in which an uniform magnetic field $\vec{B}$ exists into the plane of paper as shown in figure. Neglect any effect of electrical & gravitational forces.



1) If the actual magnitude of magnetic field is half to that of calculated in part (i) then tension in the string will be

(A) $\frac{3}{4}\frac{q^2B_0^2L}{m}$

(B) zero

(C) $\frac{q^2B_0^2L}{2m}$

(D) $\frac{2q^2B_0^2L}{m}$

2) The magnitude of the magnetic field such that no tension is developed in the string will be

(A) $\frac{B_0}{2}$

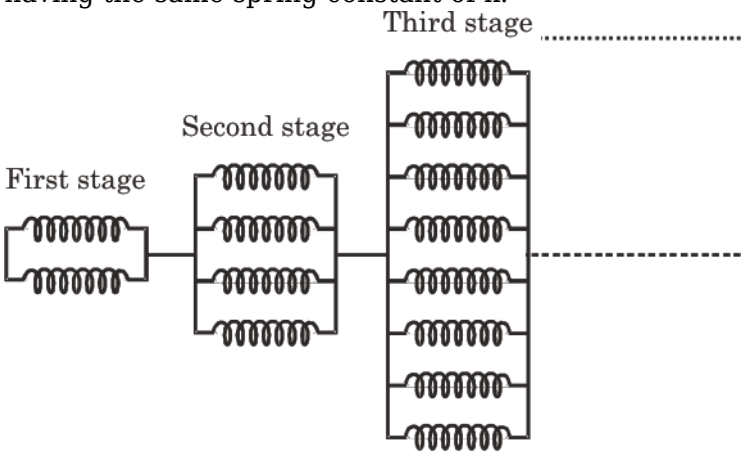
(B) B_0

(C) $2B_0$

(D) 0

Common Content for Question No. 3 to 4

A spring consists of n stages, where n tends to infinity. The first stage contains 2 springs, and the number of springs in the next stages are a geometric sequence, that is, the second stage contains 4 springs, the third stage contains 8 springs, and so on. All the springs are massless and identical having the same spring constant of k.



3) The equivalent spring constant of this spring system is :

(A) k

(B) 2k

(C) 4k

(D) 8k

4) If the total elongation of the spring system is 10cm, what is the elongation in the fourth stage, that is, the stage that contains 16 springs ?

(A) 7.5 cm

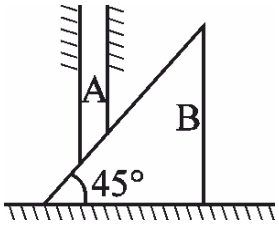
(B) 5.0 cm

(C) 2.5 cm

(D) 0.625 cm

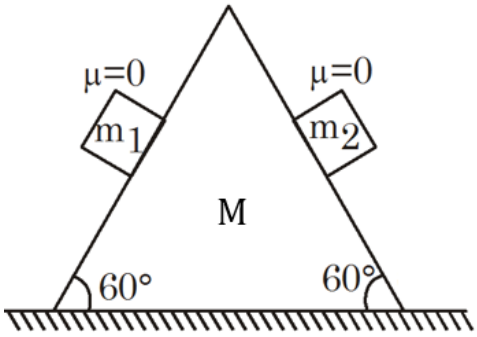
SECTION-II

1) Find the acceleration of rod A in SI units in the arrangement shown in figure if the mass of rod equal that of the wedge and the friction between all contact surfaces is negligible. Take angle of wedge as 45° & g = 9.8 m/s^2.

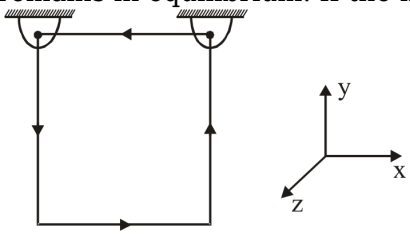


2) An empty plastic box of mass m is found to accelerate up at the rate of g/6 when placed deep inside water. The mass of sand to be filled in it so that it may accelerate down at the rate of g/6 is x_0m. The value of x_0 is.

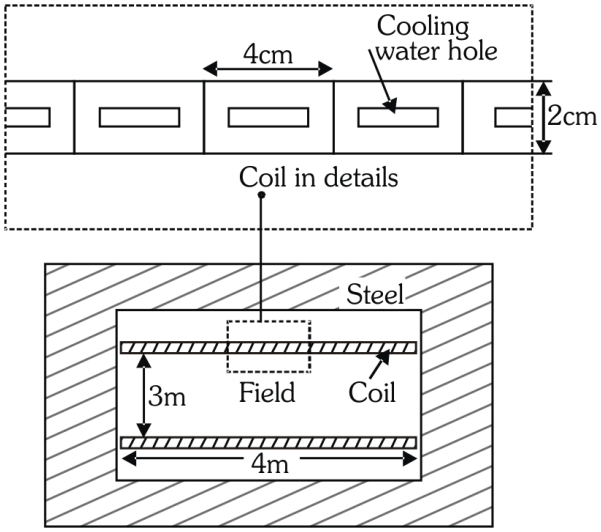
3) For what minimum value of μ_0, mass M does not move. When m_1 = 1 kg, m_2 = 2kg, M = 3/2kg. The surface between m_1 & M, m_2 & M are smooth. If $\frac{1}{\mu_0} = (\alpha)^{3/2}$ then find out the value of α. (μ^0 is coefficient of friction between M and horizontal surface) [g = 10 ms^{-2}]



4) A current carrying uniform square frame is suspended from hinged supports as shown in the figure such that it can freely rotate about its upper side. The length and mass of each side of the frame is 2m and 4kg respectively. A uniform magnetic field $\vec{B} = \left(3\hat{i} + 4\hat{j}\right)$ is applied. When the wire frame is rotated to 45° from vertical and released it remains in equilibrium. If the magnitude of current (in A) in the wire frame is I then find $\left(\frac{3}{5}\right)$I.

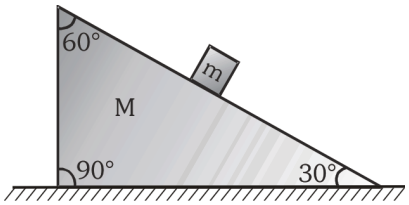


5) A large solenoid coil is made of single layer of conductor of cross section 4cm × 2cm with a cooling water hole 2cm × 1cm in the conductor. The coil, which consists of 100 turns, has a diameter of 3 meters, and a length of 4 meters. At the two ends of coil are circular steel plates to make the field uniform and to return the magnetic flux through a steel cylindrical structure to the coil, as shown in the diagram. A magnetic field of 0.25 tesla is desired. Resistivity of conductor is 3 × 10^{-8} ohm-meter. What power (in megawatt) must be supplied to provide the desired field ?

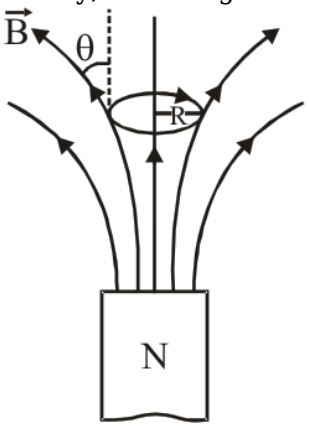


6)

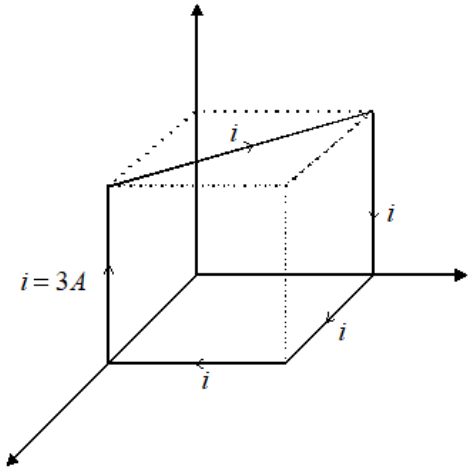
A triangular block of mass M with angle 30°, 60°, 90° rests with its 30° – 90° side on a horizontal smooth fixed table. A cubical block of mass m rests on the 60° – 30° side of the tringular block. What horizontal acceleration (in m/s^2) a must M have relative to the stationary table so that m remains stationary with respect to the triangular block- [M = 9 kg, m = 1 kg, $\sqrt{3}$ = 1.73, g = 9.8 m/s^2]



7) A superconducting current carrying ring of mass m = 40 gm and R = 2cm floats in the magnetic field of a magnet. If the field lines make an angle θ = 30° with vertical and the ring is placed co-axially, assuming i = 10A, find the magnetic field $\vec{B}$. If answer is $\frac{n}{3\pi}$, write the value of n.



8) A wire carrying 3A current is wrapped around a non - conducting cube of side 1m as shown below. Find the magnitude of net magnetic moment due to the loop (in S.I. unit).



PART-2 : CHEMISTRY

SECTION-I (i)

1)

Select the correct statement -

(A) C–O bond length of CO molecule is higher than the C–O bond length of CO_2

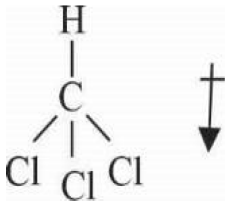
(B) C–O bond length of CO molecule is lower than the C–O bond length of CO_2

(C) N–O bond length of $NO_3^{\ominus}$ ion is higher than the N–O bond length of NO_2^{θ}

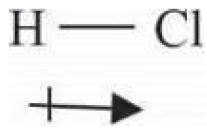
(D) N–O bond length of $NO_3^{\ominus}$ ion is lower than the N–O bond length of NO_2^{θ}

2) Which one of the following compounds indicate correct direction of dipole moment?

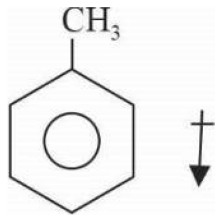
(A)

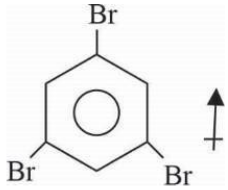


(B)



(C)



(D) 

3) Which of the following order is **CORRECT** ?

(A) Bond energy : $N_2 > O_2 > C_2 > F_2 > B_2$

(B) Bond length : $B_2 > F_2 > C_2 > O_2 > N_2$

(C) Solubility : $NaHCO_3 < KHCO_3 < RbHCO_3 < CsHCO_3$

(D) Polarization : ZnS > CdS > HgS

4) According to the Arrhenius equation,

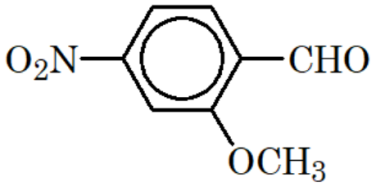
(A) A high activation energy usually implies a fast reaction

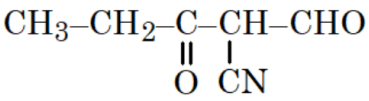
(B) Rate constant increases with an increase in temperature. This is due to a greater number of collisions whose energy exceeds the activation energy

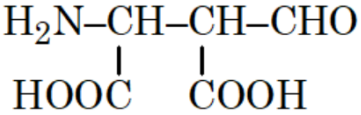
(C) Higher the magnitude of activation energy, the stronger is the temperature dependence of the rate constant.

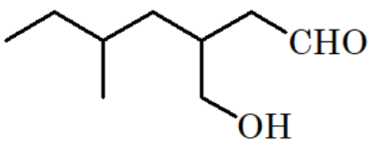
(D) The pre-exponential factor is a measure of the rate at which collisions occur, irrespective of their energy.

5) Identify in which molecule(s) –CHO is the top most functionality

(A) 

(B) 

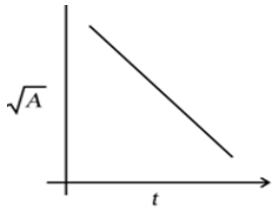
(C) 

(D) 

6) For the reaction $A \rightarrow B$, the rate law expression is $-\frac{d[A]}{dt} = k[A]^{1/2}$. If initial concentration of [A] is $[A]_0$, then

The integrated rate expression is

(A) $k = \frac{2}{t}\left(A_0^{1/2} - A^{1/2}\right)$

(B) 

The graph of $\sqrt{A}$ Vs t will be

(C) The half life period $t_{1/2} = \frac{k}{2[A]_0^{1/2}}$

(D) The time taken for 75% completion of reaction $t_{3/4} = \frac{\sqrt{[A]_0}}{k}$

SECTION-I (ii)

Common Content for Question No. 1 to 2

Dependence of rate constant of a reaction is given by Arrhenius equation $k = Ae^{-E_a/RT}$ but originally it's derived from it's differential equation : $\frac{d(\ell n\, k)}{dT} = \frac{E_a}{RT^2}$.

1) For a reaction, the dependence of rate constant (sec^{-1}) on temperature (K) is given as

$\ln k = 10 - \frac{2500}{T} + 3 \ln T$

The activation energy of reaction is

(A) 2500 + 3T

(B) 2500 R

(C) (2500 + 3T)R

(D) (2500 – 3ln T)R

Common Content for Question No. 2 to 3

For the first order dissociation reaction

$A_2B_3(aq.) \rightarrow 2A^{3+}(aq.) + 3B^{2-}(aq.)$

a definite volume of solution is taken at different instant and its osmotic pressure rise is measured. If osmotic pressure at t = 0 is 2 mm of Hg and at t = 10 min is 6 mm of Hg .

[Given density of solution is constant and equal to 1gm/ml]

2) Calculate time for 87.5 % of the reaction to complete

(A) 15 min

(B) 20 min

(C) 30 min

(D) 40 min

Common Content for Question No. 3 to 4

As one moves from sp hybridisation to sp^3. % of s-character in hybrid orbital decreases from 50% to 25% and p-character increases from 50% to 75% and in any hybrid orbital total % of s & p- character

remains 100%. By increasing p-character the hybrid orbitals become elongated hence, their overlapping extent decreases that is results into weak bond energy also bond angle decreases.

3) The molecule having bond pair in which % of s character is minimum for the central atom's hybrid orbital.

(A) SiF_4

(B) BCl_3

(C) CH_3^+

(D) CO_2

Common Content for Question No. 4 to 5

Covalent bonds are formed when two valence orbitals belonging to two different atoms overlap on each other. The electron density in the area between the two bonding atoms increases as a result of this overlapping, thereby increasing the stability of the resulting molecule.

4)

If the internuclear axis is Z then which of the following combination do not overlap.

(A) $d_{xy} + d_{xy}$

(B) $p_z + d_{z^2}$

(C) $S + p_z$

(D) $S + p_x$

SECTION-II

1) Consider the following covalent compounds in their solid state and find the value of expression (X + Y + Z).

N_2O_5, Cl_2O_6, PCl_5, XeF_6, PBr_5

Where,

X = Total number of compounds in which central atom of cationic or anionic part is sp^3 hybridised.

Y = Total number of compounds having atleast one 90° bond angle either in cationic or in anionic part

Z = Total number of compounds having atleast one $109^0 28'$ bond angle either in cationic or in anionic part

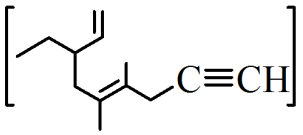
2) Bond distance in HF is 9.17×10^{-11}m. Dipole moment of HF is 6.104×10^{-30} Cm.

The percent ionic character in HF will be : (electron charge = 1.60×10^{-19}C)

3) A substance A is mixed with equal quantities of the substance B and C. At the end of 1000 second, half of the amount of A has reacted. What fraction of A will be left unreacted at the end of 2000 second. If the reaction is zero order with respect to A?

4) Catalyst A reduces the activation energy for a reaction by 10 kJ mol^{-1} at 300 K. The ratio of

rate constants, $\frac{k_T, \text{Catalysed}}{k_T, \text{Uncatalysed}}$ is e^x. The value of x is _____. [nearest integer] [Assume that the pre-exponential factor is same in both the cases.Given R = 8.31 J K^{-1} mol^{-1}]

5) In the given structure :  If (x) = total number of (- atom/s) present in principal chain.

(y) = total number of substituent (s)

(z) = total number of sp^2 and sp orbital(s)]

then, the value of $\frac{xz}{y}$ is :

6) Number of carbon(s) in 1st member of homologous series of ester is :

7) How many of the following species have linear shape ? CO_2, BF_3, I_3^{-1}, XeF_2, $HgCl_2$, NO_2^+

8) Consider the following reversible reaction,

A(g) + B(g) ⇌ C(g)

The activation energy of the backward reaction exceeds that of the forward reaction by RT (in J mol^{-1}). If the pre-exponential factor of the forward reaction is 2 times that of the reverse reaction, the absolute value of ΔG^θ (in J mol^{-1}) for the reaction at 300 K is.

(Given : ln (2) = 0.7, RT = 2500 J mol^{-1} at 300 K)

PART-3 : MATHEMATICS

SECTION-I (i)

1) Let $f(x) = \begin{cases} x^2 + a, & 0 \leq x < 1 \\ 2x + b, & 1 \leq x \leq 2 \end{cases}$ and $g(x) = \begin{cases} 3x + b, & 0 \leq x < 1 \\ x^3, & 1 \leq x \leq 2 \end{cases}$ if derivative of f(x) with respect to g(x) at x = 1 exists and is equal to k, then

(A) $a + b = -3$

(B) $a - b = 1$

(C) $\frac{ab}{k} = 3$

(D) $\frac{-b}{k} = 3$

2) Let f be a function given by $f(x)=\begin{cases}\dfrac{1}{x\ln 2}-\dfrac{1}{2^x-1}; & x\neq 0\\ \dfrac{1}{2} & ; \ x=0\end{cases}$. Then,

(A) f is continuous on R

(B) f is differentiable on R and $f'(0)$ equals $-\dfrac{\ln 2}{12}$

(C) f is not differentiable at x = 0

(D) f is differentiable on R and $f'(0)$ equals $-\dfrac{\ln 2}{6}$

3) Let [x] denote the greatest integer less than or equal to x.
If $f(x) = [x \sin \pi x]$, then f(x) is

(A) Continuous at x = 0

(B) Continuous in (–1, 0)

(C) Differentiable at x = 1

(D) Differentiable in (–1, 1)

4) For non-negative integers s and r, let $\binom{s}{r}=\begin{cases}\frac{s!}{r!(s-r)!} & \text{if } r\leqslant s\\ 0 & \text{if } r>s\end{cases}$ For positive integers m and n, let (m,n) = $\sum_{p=0}^{m+n}\dfrac{f(m,n,p)}{\binom{n+p}{p}}$ where for any non-negative integer p,

$$f(m,n,p)=\sum_{i=0}^{p}\binom{m}{i}\binom{n+i}{p}\binom{p+n}{p-i}$$

Then which of the following statements is/are TRUE?

(A) f(m, n) = g(n, m) for all positive integers m, n

(B) f(m, n + 1) = g(m + 1, n) for all positive integers m, n

(C) f(2m, 2n) = 2g(m, n) for all positive integers m, n

(D) f(2m, 2n) = (g(m, n))2 for all positive integers m, n

5) If n be a positive integer. In the expansion of $(x^2 + 2x + 2)^n$, the coefficient of

(A) x is $n.2^n$

(B) x^2 is $n^2.2^{n-1}$

(C) x^3 is ${}^{n+1}C_3 2^n$

(D) x^3 is ${}^{n+1}C_2 2^{n+3}$

6) If $f(x) = \lim_{n\to\infty} (\sin x)^{2n}$, then f is

(A) Continuous at $x = \pi$

(B) Discontinuous at $x = \frac{\pi}{2}$

(C) Discontinuous at $x = \frac{-\pi}{2}$

(D) Discontinuous at an infinite number of points.

SECTION-I (ii)

Common Content for Question No. 1 to 2

Let $f(x) = \lim_{n\to\infty} \frac{\ell n x + x^n}{(x-1)+x^n}\ (x > 0)$.

On the basis of above information, answer the following questions

1) The number of solutions of the equation $f(x) = \frac{1}{(x-1)^2}$ is

(A) 0

(B) 1

(C) 2

(D) 3

Common Content for Question No. 2 to 3

Let $f(x) = \lim_{n\to\infty} \frac{\ell n x + x^n}{(x-1)+x^n}\ (x > 0)$.

On the basis of above information, answer the following questions

2) For $f(x)$ which of the following statement is true

(A) $f(x)$ is discontinuous at x = 1

(B) non-negative difference between LHL & RHL at x = 1 is equal to 2.

(C) $f(x)$ is many-one function

(D) $f(x)$ is one-one function

Common Content for Question No. 3 to 4

Let f be a differentiable function satisfying the relation $f(x + y) = f(x) + f(y) - 2xy + (e^x - 1)(e^y - 1)$ $\forall x, y \in R$ and $f'(0) = 1$.

3) Let $g(x) = f(x) + x^2 - 2$. If the equation $|g(|x|)| = k$ has four distinct solutions then the set of values of k is

(A) (0, 1)

(B) (–2, 2)

(C) (0, 3)

(D) (0, 2)

Common Content for Question No. 4 to 5

Let f be a differentiable function satisfying the relation $f(x + y) = f(x) + f(y) - 2xy + (e^x - 1)(e^y - 1)$ $\forall x, y \in R$ and $f'(0) = 1$.

4) The value of $\lim_{x\to 0} \frac{f(2x) + 4x^2 - 2x}{x^2}$ is

(A) 2

(B) 4

(C) 8

(D) 16

SECTION-II

1) If $S_n = \sum_{r=1}^{n} \sum_{t=0}^{r-1} \left(\frac{1}{6^n} {}^nC_r \, {}^rC_t \, 4^t\right)$, then value $(1 - S_n)^{-\frac{1}{n}}$ is equals to

2) If $S_n = \sum_{r=1}^{n} \sum_{t=0}^{r-1} \left(\frac{1}{6^n} {}^nC_r \, {}^rC_t \, 4^t\right)$, then value of □ where □ = $\frac{1}{2}\sum_{n=1}^{\infty}(1 - S_n)$ is

3) The number of real value(s) of 'a' for which the three numbers $4^a, 4^{a^2}$ and 4^{a^3} are three consecutive terms of a non constant A.P., is equal to

4) The value of $\lim_{x\to \frac{\pi}{2}} (\tan x)^{\cot x} \cdot (\sec x)^{\cos x}$ is

5) If $f(x) = \begin{cases} \frac{\sin^{-1}(\{x\}^2 + \{x\}^3) - x}{\{x\}}, & x \neq 0 \\ -1, & x = 0 \end{cases}$, then the absolute value of RHD of $f(x)$ at $x = 0$, is (where $\{.\}$ denotes fractional part function)

6)

If $f(x) + f(y) = f\left(\frac{x + y}{1 + xy}\right)$ for all $x, y \in R$ $(xy \neq -1)$ and $\lim_{x\to 0} \frac{f(x)}{x} = 3$, then $f'\left(\frac{1}{2}\right)$ is equal to

7)

Let $f(x) = |\sin x||(x - 1)||(x - \pi)||(x - 2\pi)|\left|x - \frac{\pi}{2}\right|$ (where $\frac{9}{10} \leq x \leq 2\pi$) and $g(x) =$ □n(x – 1) + |x – 2| + |x

+ 3| (where x > 1). Let number of points where $f(x)$ and $g(x)$ are non differentiable are respectively λ_1, λ_2, then $\lambda_1 + \lambda_2$ is

8) Let $f(x) = \begin{cases} \tan^{-1} x, & ; \ |x| \leq 1 \\ \frac{\pi}{4}\operatorname{sgn}(x) + \frac{x-1}{2} & ; \ |x| > 1 \end{cases}$, (where sgn (x) denotes signum function)
Then the value of $f_+'(1)$ is equal to _____.

ANSWER KEYS

PART-1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4	5	6
A.	A,B,C,D	A,D	A,B,C,D	B,C	A,C,D	A,B,C

SECTION-I (ii)

Q.	7	8	9	10
A.	C	B	A	D

SECTION-II

Q.	11	12	13	14	15	16	17	18
A.	4.90	0.40	3.00	6.00	2.98 to 3.00	5.65 to 5.66	6.00	4.50

PART-2 : CHEMISTRY

SECTION-I (i)

Q.	19	20	21	22	23	24
A.	B,C	A,B,C	B,C	B,C,D	A,D	A,B,D

SECTION-I (ii)

Q.	25	26	27	28
A.	C	C	A	D

SECTION-II

Q.	29	30	31	32	33	34	35	36
A.	7.00	41.58 to 41.62	0.00	4.00	48.00	2.00	5.00	4250.00

PART-3 : MATHEMATICS

SECTION-I (i)

Q.	37	38	39	40	41	42
A.	A,B,C,D	A,B	A,B,D	A,B,D	A,B,C	A,B,C,D

SECTION-I (ii)

Q.	43	44	45	46
A.	C	C	D	A

SECTION-II

Q.	47	48	49	50	51	52	53	54
A.	**1.20**	**2.5**	**0.00**	**1.00**	**1.00**	**4.00**	**3.00**	**0.50**

SOLUTIONS

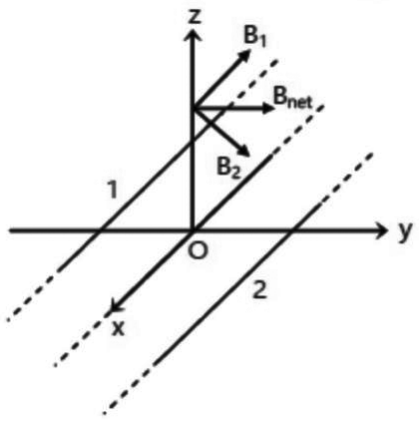
PART-1 : PHYSICS

1) (A) $B = \frac{\mu_0 I}{2\pi a} - \frac{\mu_0 I}{2\pi a} = 0$

(B) On y-axis say at (y, 0, 0)

$$B = \frac{-\mu_0 I}{2\pi(a+y)}\hat{k} + \frac{\mu_0 I}{2\pi(a-y)}\hat{k}$$

So except at origin, B has only z - components



(C)

B cannot has x-component as B is perpendicular to direction of I.

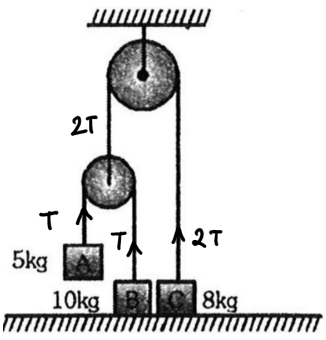
2) $\sigma = \frac{\theta}{\pi R^2}$

$dI = dqf = (\sigma 2\pi r du)\frac{\omega}{2\pi}$

$$dB = \int_0^R \frac{\mu_0 dI}{2r}$$

$$B = \frac{\mu_0 \theta \omega}{2\pi R}$$

3)



Accelerations of blocks are related as

$a_A - a_B = 2a_4$

For 5 kg : $5g - T = 5\,a_A$
For 10 kg : $T - 10g = 10\,a_B$
For 8kg : $2T - 8g = 8\,a_C$
Solving for T, $T = 5g < 10g \Rightarrow a_B = 0$
Solving for a_A & a_C, $a_A = \frac{g}{7} m/s^2$ $a_C = \frac{g}{14} m/s^2$

4)

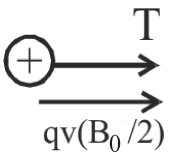
The block A will just starts moving up if,
$mg \sin\theta_1 + f = mg \sin\theta_2$
$\Rightarrow mg \sin\theta_1 + \mu mg \cos\theta_1 = mg \sin\theta_2$
$\Rightarrow \mu = \frac{\sin\theta_2 - \sin\theta_1}{\cos\theta_1}$
When block A moves upwards
$f = mg \sin\theta_2 - mg \sin\theta_1 > 0$
$\sin\theta_2 > \sin\theta_1 \Rightarrow \theta_2 > \theta_1$

5) $R = \ell = \frac{mv}{qB}$
$v = \frac{qB\ell}{m}$
$v > \frac{qB\ell}{m}$
$\therefore T = \frac{2\pi m}{qB}$
Independent of speed.

6)

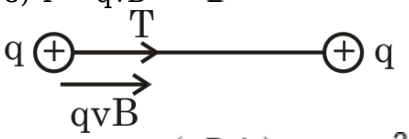
(A) $a = \frac{0.2g}{0.7} = \frac{2g}{7}$
(B) $0 = -7 + \frac{2g}{7}t$
t = 2.5 sec.
(C) in 2.5 sec. $S_1 = 7 \times 2.5 - \frac{1}{2} \times \frac{2g}{7} \times (2.5)^2$
in next 2.5 sec. $S_2 = 0 \times 2.5 + \frac{1}{2} \times \frac{2g}{7} \times (2.5)^2$
in 5 sec. $S_1 + S_2 = 17.5$ m.
(D) $V = -7 + \left(\frac{2g}{7} \times 5\right)$
$V \neq -7 m/s$

7) $T + \frac{qvB_0}{2} = \frac{mv^2}{L}$



$$T = \frac{m\left(\frac{qB_0L}{m}\right)^2}{L} - q\left(\frac{qB_0L}{m}\right) \times \frac{B_0}{2} = \frac{q^2B_0^2L}{2m}$$

8) $T = qvB = \frac{mv^2}{L}$



$T = 0; \quad \Rightarrow q\left(\frac{qB_0L}{m}\right)B = \frac{mv^2}{L}$

$\Rightarrow B = B_0$

9) $\frac{1}{k_{eq}} = \frac{1}{2k} + \frac{1}{4k} + \frac{1}{8k}$

$\frac{1}{k_{eq}} = \frac{1}{2k} \cdot \frac{1}{1-\frac{1}{2}} = \frac{1}{k}$

$k_{eq} = k$

10) $F = k_{eq} \times 10 = 10k$

Force will be same for every group of spring at every stage.

Therefor $10k = 16kx$

$\Rightarrow x = \frac{10}{16} = \frac{5}{8} = 0.625$

11) $N \sin \theta = ma$ ..... (i)

$mg - N \cos \theta = mb$ ..... (ii)

Acceleration along perpendicular to contact surface must be same

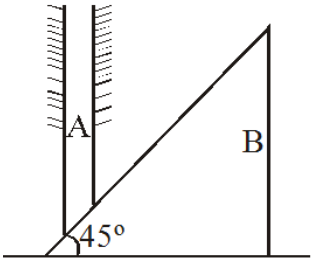
$a \cos 45^\circ = b \sin 45^\circ$

$a = b$ ..... (iii)

put (ii) in (ii)

$N \sin 45^\circ = ma$

$\frac{N}{\sqrt{2}} = ma$



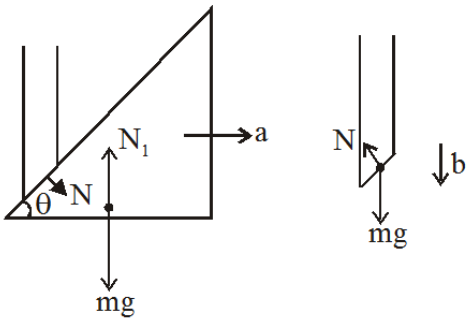
$mg - N \cos 45^\circ = ma$

$mg - \frac{N}{\sqrt{2}} = ma$

$mg - ma = ma$

$a = \frac{g}{2}$

$a = b = \frac{g}{2}$

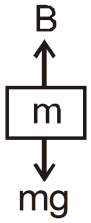


12)

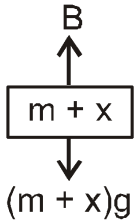
$B - mg = \frac{mg}{6}$

$B = \frac{7}{6} mg.$

$(m + x)g - B = (m + x)\frac{g}{6}$

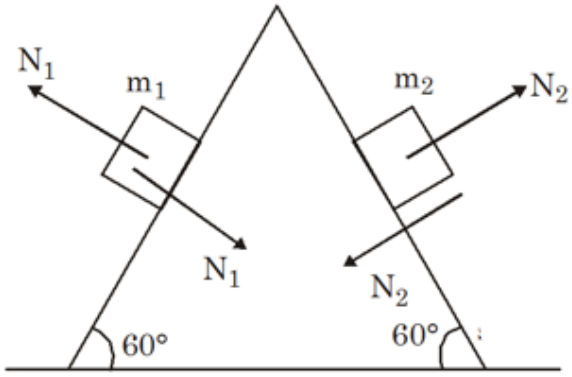


$6(m + x)g - 6B = (m + x)g$



$5m + 5x = 7m$

$x = \frac{2}{5} m.$



Net horizontal force a bigger block

$(N_2 - N_1)\sin\theta = \mu N_{net}$

$N_{net} = (N_1 + N_2)\cos\theta + Mg$

$N_1 = m_1 g\cos\theta$

13)

14) $\vec{\mu}$ (Magnetic moment of loop) when it is lifted by 45° = $i\ell^2\left(\frac{\hat{j}+\hat{k}}{\sqrt{2}}\right)$

$\vec{\tau}$ due to magnetic field

$= \vec{\mu} \times \vec{B} = \frac{i\ell^2}{\sqrt{2}}\left[\left(\hat{j}+\hat{k}\right) \times \left(3\hat{i}+4\hat{j}\right)\right]$

$\vec{\tau}$ due to mg (about top edge)

$= 4mg\frac{\ell}{2}\cos 45^\circ\hat{i}$

For equilibrium net torque along

X-axis = 0

$\frac{4mg\ell}{2\sqrt{2}} = \frac{4i\ell^2}{\sqrt{2}} \Rightarrow i = \frac{mg}{2\ell} = 10A$

15) $B = \ell_0\frac{NI}{L}$

$\Rightarrow I = \frac{BL}{\mu_0 N} = \frac{0.25 \times 4}{4\pi \times 10^{-7} \times 100} = 7960\,A$

$R = P\frac{L}{A}$

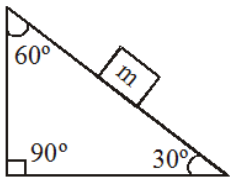
$\Rightarrow R = 3 \times 10^{-8} \times \frac{(100 \times 2\pi \times 1.5)}{(4 \times 2 - 2 \times 1) \times 10^{-4}} = 0.0471\,\Omega$

$P = I^2R \Rightarrow P = (7960)^2 \times 0.0471$

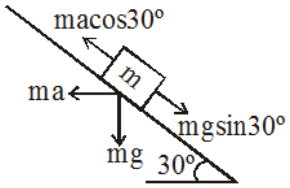
16) M = 9kg

m = 1kg

g = 9.8 m/s^2



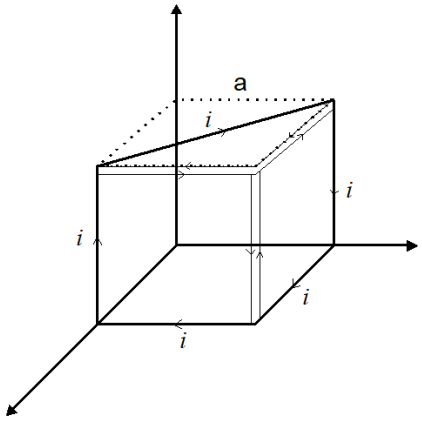
Let acceleration 'a' is given to 'M'
So block 'm' get for "ma" as



for constant velocity of zero of block 'm'
ma cos 30º = mg sin 30º
a = g tan 30º

$\boxed{a = 5.658\ m/s^2}$

17) mg = [I 2πR B] sin 30º



18)

Here $\vec{M} = -ia^2\hat{i} - i\frac{a^2}{2}\hat{j} - ia^2\hat{k}$

$\vec{M} = -ia^2\left(\hat{i} + \frac{\hat{j}}{2} + \hat{k}\right) \Rightarrow \left|\vec{M}\right| = \frac{3}{2}ia^2$

$= \frac{3}{2} \times 3 \times 1\left|\vec{M}\right| = \frac{9}{2}$ amphere - meter^2

PART-2 : CHEMISTRY

(I)

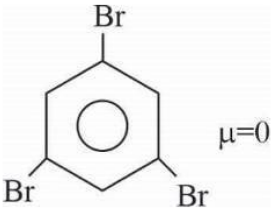
	$^{\ominus}C \equiv O^{\oplus}$		$O = C = O$
B.O.	3	>	2
B.L.(C-O)	CO	<	CO_2

(II)



B.O.	$\frac{3}{2}$	>	$\frac{4}{3}$
B.L. (N-O)	$NO_2^{\ominus}$	<	$NO_3^{\ominus}$

19)

20) 

21)

B.E. : $N_2 > C_2 > O_2 > B_2 > F_2$

Interelectronic repulsion

(A)

(B) Based on B.O and atomic size

(D) HgS > CdS > ZnS : Poor shielding by d and f-orbital

22) According to Arrhenius Equation

$k = Ae^{\frac{-Ea}{RT}}$

Therefore at $T \rightarrow 0$

$e^{-Ea/RT} \rightarrow 0$

Therefore rate constant decreases with temperature.

And as $T \rightarrow \infty\ k \rightarrow A.$

Thus, as temperature increases, the rate constant increases and approaches a value close to the Arrhenius constant.

Now

$$\frac{k}{dT} = -\frac{E_a A}{RT} \times e^{-\frac{E_a}{RT}}$$

Hence,

Higher the activation energy higher is the required temperature for reaction to occur, and hence greater is the temperature dependency of the rate constant.

23)

Functionality priority among given groups is
–COOH > –CN > –CHO > –CO– > –OH > $–NH_2$ > $–NO_2$ and $–OCH_3$.

24)

$A \rightarrow B$

$$\frac{-dA}{dt} = k[A]^{1/2}$$

$$\int \frac{dA}{(A)^{1/2}} = -k \int dt$$

$2(A)^{1/2} = -kt + C$

Condition

$A = A_0$ at $t = 0$

$$c = 2[A]_0^{1/2}$$

$$2[A]^{1/2} = -kt + 2[A_0]^{1/2}$$

$$k = \frac{2}{t}\left[[A_0]^{1/2} - [A]^{1/2}\right]$$

$$t_{1/2} = \frac{[A_0]^{1/2}}{k} \times \left(2 - \sqrt{2}\right)$$

Now,

$$t \rightarrow t_{3/4}, A \rightarrow \frac{A_o}{4}$$

$$k = \frac{2}{t_{3/4}}\left[A_o^{1/2} - \frac{A_o^{1/2}}{2}\right]$$

$$t_{3/4} = \frac{A_o^{1/2}}{k}$$

25) $\frac{d}{dT}(\ln k) = \frac{E_a}{RT^2}$

$$0 + \frac{2500}{T^2} + \frac{3}{T} = \frac{E_a}{RT^2} \quad \Rightarrow E_a = (2500 + 3T)R$$

26) $k \times 10 = \ln 2$

$$k = \left(\frac{\ln 2}{10}\right) = \frac{\ln 2}{t_{1/2}}$$

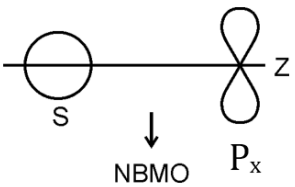
$\Rightarrow t_{1/2} = 10$ min

$t_{0.875} = 3 \times 10 = 30$ min

27) SiF_4 $\Rightarrow sp^3$ 25% s character

BCl_3 & CH_3^+ $\Rightarrow sp^2$ 33.33% s character

CO_2 $\Rightarrow$ sp 50% s character

28) 

29) $N_2O_5(s) \rightarrow [NO_2]^+[NO_3^-]$
$Cl_2O_6(s) \rightarrow [ClO_2]^+[ClO_4^-]$
$PCl_6(s) \rightarrow [PCl_4]^+[PCl_6^-]$
$XeF_6(s) \rightarrow [XeF_5]^+[F^-]$
$PBr_5(s) \rightarrow [PBr_4]^+[Br^-]$
X = 3, Y = 1, Z = 3

30)

Bond distance (d) = 9.17×10^{-11} m
μ (actual) = 6.104×10^{-30} C m
electron charge = 1.6×10^{-19} C

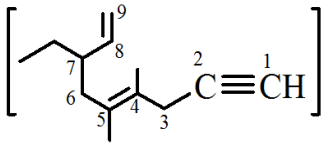
$$\%\text{Ionic character} = \frac{\mu(\text{actual})}{\mu(\text{calculated})} \times 100$$
$$= \frac{6.104 \times 10^{-30}}{9.17 \times 10^{-11} \times 1.6 \times 10^{-19}} \times 100$$
$$= 41.60$$

31) Reaction zero order w.r.t to A then, when half of A reached

	A + B + C ⟶ Product
at t = 0	a_0 a_0 a_0
at t = 1000 sec.	$\frac{a_0}{2}$ $\frac{a_0}{2}$ $\frac{a_0}{2}$

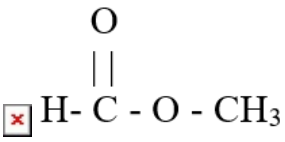
Then we know for zero orders Reaction $(a_0 - a_t) = kt$
given $a_t = \frac{a_0}{2}$.
Then $k \times t = \frac{a_0}{2}$
$k = \frac{a_0}{2 \times t} = \frac{a_0}{2 \times 1000} = \frac{a_0}{2000}$
at t=2000sec.
$a_0 - a_t = \frac{a_0}{2000} \times 2000$
a_t=0 zero fraction left.

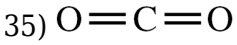
32) $k = Ae^{\frac{-Ea}{RT}}$
$k_{cat} = Ae^{-\frac{Ea^1}{RT}}, k_{uncat} = Ae^{-\frac{Ea}{RT}}$
$\frac{k_{cat}}{k_{uncat}} = e^{\frac{Ea - Ea^1}{RT}} = e^{\frac{10 \times 1000}{8.31 \times 300}} = e^{4.009} = e^x$
⇒ x = 4

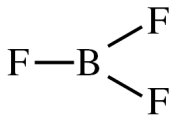


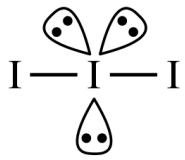
33)
(x) = 9
(y) = 3
(z) = 16
Here 4 sp^2 Hybridised c-atoms and 2sp hybridised c-atoms are present and each sp^2 and sp hybridised carbon contains 3 and 2 hybridised orbitals respectively.

34)

(2) 

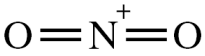
35)  Linear

 Trigonal planar

 Linear

 Linear

 Linear

 Linear

Ans. 5.00

36) $\Delta H = E_{a_f} - E_{a_b} = -RT$
$E_b = E_f + RT$
$A_f = 2A_b$
$K = 2 \times e^{RT/RT} = 2e$
$\Delta G = \Delta G° + RT \ln K$
$\Delta G° = -RT \ln K$
$= -RT \ln(2e) = -RT(\ln 2 + \ln e)$
$= -2500\ (0.7 + 1)$
$= -4250$
or, $\Delta G° = -4250\ J\ mol^{-1}$

PART-3 : MATHEMATICS

37)

If it is differentiable
then 1+ a = 2+b ⇒ a – b = 1
& 3 + b = 1 ⇒ b = –2
a = –1
(B) a + b = –3

$$\frac{f'(x)}{g'(x)} = \frac{2}{3} = k$$

(C) $$\frac{ab}{k} = \frac{(-1)(-2)}{2/3} = 3$$

(D) $$-\frac{b}{k} = \frac{(2)}{2/3} = 3$$

$$\lim_{x\to 0}\left(\frac{1}{x\ln 2} - \frac{1}{2^x - 1}\right)$$

$$= \frac{1}{\ln 2}\lim_{x\to 0}\frac{2^x - 1 - x\ln 2}{x\left(2^x - 1\right)}$$

$$= \frac{1}{(\ln 2)^2}\lim_{x\to 0}\frac{2^x - 1 - x\ln 2}{x^2}$$

$$= \frac{1}{(\ln 2)^2}\lim_{x\to 0}\left(\frac{\left(1 + x\ln 2 + \frac{(x\ln 2)^2}{2!} + \cdots\right) - 1 - x\ln 2}{x^2}\right)$$

38) $$= \frac{1}{2} = f(0)$$

$$f'(0) = \lim_{x\to 0}\frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0}\frac{\left(\frac{1}{x\ln 2} - \frac{1}{2^x - 1} - \frac{1}{2}\right)}{x}$$

$$= \frac{1}{2\ln 2}\lim_{x\to 0}\frac{2\left(2^x - 1\right) - 2x\ln 2 - x\ln 2\left(2^x - 1\right)}{x^2\left(2^x - 1\right)} \ldots (1)$$

Now, $2^x = e^{x\ln 2} = 1 + x\ln 2 + \frac{(x\ln 2)^2}{2!} + \frac{(x\ln 2)^3}{3!} + \cdots$

$$\Rightarrow 2^x - 1 = x\ln 2 + \frac{(x\ln 2)^2}{2!} + \frac{(x\ln 2)^3}{3!} + \ldots(2)$$

$$2\left(2^x - 1\right) = x(2\ln 2) + x^2(\ln 2)^2 + x^3\frac{(\ln 2)^3}{3}\ldots(3)$$

$$x\ln 2\left(2^x - 1\right) = (x\ln 2)^2 + x^3\frac{(\ln 2)^3}{2} + \ldots(4)$$

Using, (1), (2), (3), (4), we get

$$f'(0) = \frac{1}{2(\ln 2)^2}\cdot\left[\frac{(\ln 2)^3}{3} - \frac{(\ln 2)^3}{2}\right] = -\frac{\ln 2}{12}$$

39) By the definition of [x], we have f(x) = $[x\sin\pi x] = 0$ for $-1 \leqslant x \leqslant 1$, because $0 \leqslant x\sin\pi x \leqslant 1$.
Also, $f(x) = [x\sin\pi x] = -1$ for $1 < x < 1 + h$ for some small appropriate $h > 0$, $\sin\pi x$ because is negative and $\geqslant -1$ for $1 < x < 1 + h$.
Thus f(x) is constant and equal to 0 in the interval [–1, 1] and so it is continuous and differentiable in (–1, 1).
In particular, f(x) is continuous at x = 0 and in the interval (–1, 0).
At x = 1, $\lim_{x\to 1+} f(x) = -1$ and $\lim_{x\to 1-} f(x) = 0$.
Hence f is not continuous at x = 1 and, in particular, not differentiable at x = 1.

40)

Solving

$$f(m,n,p) = \sum_{i=0}^{p} {}^{m}C_i\ {}^{n+i}C_p.\ {}^{p+n}C_{p-i}$$

$${}^{m}C_i.\ {}^{n+i}C_p.{}^{p+n}C_{p-i}$$

$${}^{m}C_i.\frac{(n+i)!}{p!(n-p+i)!}\times\frac{(n+p)!}{(p-i)!(n+i)!}$$

$${}^{m}C_i\times\frac{(n+p)!}{p!}\times\frac{1}{(n-p+i)!(p-i)!}$$

$${}^{m}C_i\times\frac{(n+p)!}{p!n!}\times\frac{n!}{(n-p+i)!(p-i)!}$$

$${}^{m}C_i.{}^{n+p}C_p.\ {}^{n}C_{p-i}\left\{{}^{m}C_i.\ {}^{n}C_{p-i} = {}^{m+n}C_p\right\}$$

$f(m,n,p) = {}^{n+p}C_p.{}^{m+n}C_p$

$$\frac{f(m,n,p)}{{}^{n+p}C_p} = {}^{m+n}C_p$$

Now,

$$g(m,n) = \sum_{p=0}^{m+n}\frac{f(m,n,p)}{{}^{n+p}C_p}$$

$$g(m,n) = \sum_{p=0}^{m+n} {}^{m+n}C_p$$

$g(m,n) = 2^{m+n}$

(A) $g(m,n) = q(n,m)$

(B) $g(m,n+1) = 2^{m+n+1}$

$g(m+n,n) = 2^{m+1+n}$

(D) $g(2m,2n) = 2^{2m+2n}$

$= (2^{m+n})^2$

$= (g(m,n))^2$

41) $(x^2 + 2x + 2)^n = (x^2 + (2 + 2x))^n$
$= {}^nC_0 x^{2n}(2 + 2x)^0 + ...$
$.....+ {}^nC_{n-1}x^2(2 + 2x)^{n-1} + {}^nC_n(2 + 2x)^n$
for coeff. of $x = {}^nC_n.{}^nC_1.2^{n-1}.(2x)^1$
$= (n.2^n).x$
for coeff of x^2
$= {}^nC_{n-1}.x^2.{}^{n-1}C_0 2^{n-1} + {}^nC_n {}^nC_2.2^{n-2}.(2x)^2$
for coeff of $x^3 = ({}^{n+1}C_3.2^n)$

42) $\lim_{n\to\infty} x^{2n} = \begin{cases} 0 & \text{if } |x| < 1 \\ 1 & \text{if } |x| = 1 \end{cases}$

$\Rightarrow f(x) = \lim_{n\to\infty} (\sin x)^{2n} = \begin{cases} 0 & \text{if } |\sin x| < 1 \\ 1 & \text{if } |\sin x| = 1 \end{cases}$

This shows that f is continuous for all x, except possibly when |sin x| = 1, i.e., when $x = (2k+1)\frac{\pi}{2} (k \in I)$. For these points, we have

$\lim_{x\to(2k+1)\frac{\pi}{2}} f(x) = 0 \neq 1 = f\left((2k+1)\frac{\pi}{2}\right)$

Hence f(x) is discontinuous at these points.

43) For $f(x) = \frac{\ell nx}{(x-1)}$; $0 < x < 1$ $f(x) = 1$; $x = 1$

$f(x) = \lim_{x\to\infty} \frac{\frac{\ell nx}{x^n} + 1}{\frac{(x-1)}{x^n} + 1} = 1;\ x > 1$

$f(x) = \begin{cases} \frac{\ell nx}{(x-1)} & ;\ 0 < x < 1 \\ 1 & ;\ x \geqslant 1 \end{cases}$

$f(x) = \frac{1}{(x-1)^2}$

(a) for $0 < x < 1$

$$\frac{\ell nx}{(x-1)} = \frac{1}{(x-1)^2}$$

$$\ell nx = \frac{1}{(x-1)}$$

44) For $f(x) = \frac{\ell nx}{(x-1)}\ ;\ 0 < x < 1$

$f(x) = 1;\ x = 1$

$$f(x) = \lim_{x\to\infty} \frac{\frac{\ell nx}{x^n} + 1}{\frac{(x-1)}{x^n} + 1} = 1;\ x > 1$$

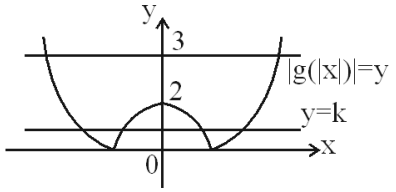
$$f(x) = \begin{cases} \frac{\ell nx}{(x-1)} & ;\ 0 < x < 1 \\ 1 & ;\ x \geqslant 1 \end{cases}$$

$f(x)$ is many one function

45) $g(x) = f(x) + x^2 - 2 = e^x - x^2 - 1 + x^2 - 2$

$g(x) = e^x - 3$

graph = $|g(|x|)|$ is



Now the $|g(|x|)| = k$ has four roots

$\therefore k \in (0, 2)$

46) $f'(x) = \lim_{h\to\infty} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h\to 0} \frac{f(x) + f(h) - 2xh + (e^x - 1)(e^h - 1) - f(x)}{h}$$

$$= \lim_{h\to 0} \left(\frac{f(h)}{h} + (e^x - 1)\frac{(e^h - 1)}{h} \right) - 2x$$

$$= f'(0) + e^x - 1 - 2x$$

$= 1 + e^x - 1 - 2x = e^x - 2x$

$\therefore\ f(x) = e^x - x^2 - 1$

$$\lim_{x\to 0} \frac{f(2x) + 4x^2 - 2x}{x^2}$$

$$= \lim_{x\to 0} \frac{e^{2x} - 4x^2 - 1 + 4x^2 - 2x}{x^2}$$

$$= \lim_{x\to 0} \frac{e^{2x} - 2x - 1}{x^2}$$

$$= \lim_{x\to 0} \frac{1 + 2x + \frac{4x^2}{2!} + .. - 2x - 1}{x^2} = 2$$

47) $S_n = \sum_{r=1}^{n}\left(\frac{1}{6^n}\,{}^nC_r\left(\sum_{t=0}^{r-1}{}^rC_t\,4^t\right)\right)$

$= \sum_{r=1}^{n}\frac{1}{6^n}\,{}^nC_r\,\left[(1+4)^r-4^r\right]$

$= \frac{1}{6^n}\sum_{r=1}^{n}\left({}^nC_r\,5^r-{}^nC_r\,4^r\right)$

$= \frac{1}{6^n}\left[(1+5)^n-1-\left((1+4)^n-1\right)\right] = \frac{6^n-5^n}{6^n} = 1-\left(\frac{5}{6}\right)^n$

$(1-S_n)^{-\frac{1}{n}} = \frac{6}{5} = 1.2$

48) $S_n = \sum_{r=1}^{n}\left(\frac{1}{6^n}\,{}^nC_r\left(\sum_{t=0}^{r-1}{}^rC_t\,4^t\right)\right)$

$= \sum_{r=1}^{n}\frac{1}{6^n}\,{}^nC_r\,\left[(1+4)^r-4^r\right]$

$= \frac{1}{6^n}\sum_{r=1}^{n}\left({}^nC_r\,5^r-{}^nC_r\,4^r\right)$

$= \frac{1}{6^n}\left[(1+5)^n-1-\left((1+4)^n-1\right)\right] = \frac{6^n-5^n}{6^n} = 1-\left(\frac{5}{6}\right)^n$

$\frac{1}{2}\sum_{n=1}^{\infty}(1-S_n) = \frac{1}{2}\sum_{n=1}^{\infty}\left(\frac{5}{6}\right)^n = \frac{1}{2}\frac{\frac{5}{6}}{1-\frac{5}{6}} = 2.5.$

49) $4^a, 4^{a^2}, 4^{a^3}$ are in A.P.

$2.4^{a^2} = 4^a + 4^{a^3}$

Now using A.M. ≥ G.M.

$2^{2a^2+1} = 2^{2a} + 2^{2a^3}$ .... (1)

$\frac{2^{2a}+2^{2a^3}}{2} \geqslant \sqrt{2^{2a}.2^{2a^3}}$

$\Rightarrow \frac{(2)^{2a^2+1}}{2} \geqslant (2)^{a+a^3}$

$\Rightarrow (2)^{2a^2} \geqslant (2)^{a+a^3}$

$\Rightarrow 2a^2 \geqslant a\left(1+a^2\right)$

$\Rightarrow a^2 + 1 - 2a \leq 0$

$\Rightarrow (a-1)^2 \leq 0$

which is not possible as $a \neq 1$

50) $y = \lim_{x\to\frac{\pi}{2}}(\tan x)^{\cot x}(\infty)^0$

$\ell n y = \cot x \,.\, \ell n \tan x = \lim_{x\to\frac{\pi}{2}}\frac{\ell n(\tan x)}{\tan x} = \frac{\sec^2 x}{(\tan x).\sec^2 x}$

ℓny = 0 ⇒ y = 1

Similarly

$y = (\sec x)^{\cos x}$

$\ell n y = (\cos x)(\ell n \sec x)$

$$\ell n y = \lim_{x \to \frac{\pi}{2}} \frac{\ell n (\sec x)}{\sec x} = \frac{\sec x \tan x}{(\sec x) . \sec x \tan x} = 0$$

$y = 1 \Rightarrow k = 1$

51)

$$f'(0^+) = \lim_{h \to 0} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \to 0} \frac{\frac{\sin^{-1}\left(h^2+h^3\right) - h}{h} + 1}{h}$$

$$= \lim_{h \to 0} \frac{\sin^{-1}\left(h^2 + h^3\right)}{h^2} \frac{\left(h^2 + h^3\right)}{\left(h^2 + h^3\right)} = 1$$

$$\lim_{x \to 0^+} f(x) = \lim_{h \to 0} \frac{\sin^{-1}\left(h^2 + h^3\right) - h}{h}$$

$$= \lim_{h \to 0} \frac{\frac{1}{\sqrt{1-\left(h^2+h^3\right)^2}} . \left(2h + 3h^2\right) - 1}{1} = 1$$

$$f(x) + f(y) = f\left(\frac{x+y}{1+xy}\right)$$

$$\text{put } x = y = 0 \Rightarrow f(0) = 0$$

$$\text{put } y = -x \Rightarrow f(x) = -f(-x)$$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{f(x+h) + f(-x)}{h}$$

52)

$$= \lim_{h\to 0} \frac{f\left(\frac{h}{1-x^2-xh}\right)}{\left(\frac{h}{1-x^2-xh}\right)} \lim_{h\to 0} \frac{1}{1-x^2-xh}$$

$$f'(x) = \frac{3}{1-x^2}$$

$$\Rightarrow f'\left(\frac{1}{2}\right) = 4$$

53)

$f(x)$ is not differentiable at $x = 1, \frac{\pi}{2}$ and $g(x)$ is non differentiable at $x = 2$. Total number of points are three.

54) $f_+^1(1) = \lim_{h\to 0} \frac{f(1+h)-f(1)}{h}$

$= \lim_{h\to 0} \frac{\frac{\pi}{4} + \frac{1+h-1}{2} - \frac{\pi}{4}}{h} = \frac{1}{2}$